



Database Systems

Final Project

SOCIAL MEDIA DATABASE SYSTEM

A relational backend covering user management, content sharing, social graphs, messaging, and notifications — implemented in MySQL and queried end-to-end.

SECTION INSTRUCTOR

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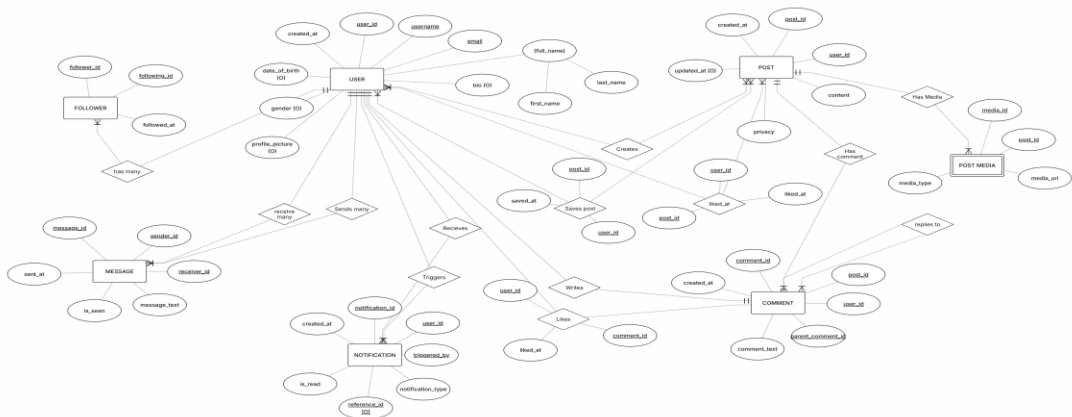
Project Overview

This project implements the backend relational database for a social media platform — covering user management, content sharing, social interactions, messaging, and notifications. The design follows the complete database lifecycle: requirement analysis, ER modeling, normalization, and finally the SQL layer documented in this report.

Core Entities

- 01 Users** — Central entity storing profile details — username, email, bio, profile picture, gender, and date of birth.
- 02 Posts** — User-created content with privacy settings and creation/update timestamps.
- 03 Post Media** — Image and video attachments linked to a post (one post, many media items).
- 04 Comments** — User comments on posts, with self-referencing keys for nested/threaded replies.
- 05 Followers** — Junction entity capturing follow relationships between users (many-to-many).
- 06 Messages** — Direct messages exchanged between two users, with read/seen tracking.
- 07 Saved Posts** — Junction entity linking users to the posts they've bookmarked.
- 08 Post Likes** — Junction entity tracking which users liked which posts, with a timestamp.
- 09 Comment Likes** — Junction entity tracking which users liked which comments.
- 10 Notifications** — System alerts triggered by likes, comments, follows, and messages.

Entity-Relationship Diagram



01

JOINS

1.1 (a) INNER JOIN — Posts with their author info

```
288 --
289 -- 3.1 JOINS
290 --
291
292 -- (a) INNER JOIN — Posts with their author info
293 • SELECT p.post_id, u.username, u.full_name, p.content, p.privacy, p.created_at
294 FROM Posts p
295 INNER JOIN Users u ON p.user_id = u.user_id
296 ORDER BY p.created_at DESC;
297
```

post_id	username	full_name	content	privacy	created_at
1	hussain_d	Hussain Durrani	Just finished implementing BFS in C++	Public	2026-06-15 18:23:52
9	hussain_d	Hussain Durrani	DSA assignment submitted on time for once	Public	2026-06-15 18:23:52
6	talha_k	Talha Khurram	Just deployed my portfolio website. Check it out!	Public	2026-06-15 18:23:52
14	talha_k	Talha Khurram	Anyone else obsessed with Notion for productiv...	Public	2026-06-15 18:23:52
7	ali_h	Ali Hamza	Missing home-cooked food ☹️ Lahore ki biryani ...	Friends	2026-06-15 18:23:52
2	sara_m	Sara Malik	Beautiful sunset from Lahore today 🌅	Public	2026-06-15 18:23:52
8	sara_m	Sara Malik	Photography tip: golden hour is everything 📸	Public	2026-06-15 18:23:52
3	haseeb_r	Haseeb Rehman	New personal best at the gym 💪	Public	2026-06-15 18:23:52
15	haseeb_r	Haseeb Rehman	Just hit 500 followers! Thank you all 🙏	Public	2026-06-15 18:23:52
11	zara_a	Zara Ahmed	Eid Mubarak to everyone! 🌙	Public	2026-06-15 18:23:52
10	usman_b	Usman Baig	Anime recommendation: Vinland Saga — master...	Public	2026-06-15 18:23:52
4	aisha_n	Aisha Nadeem	Started reading "Clean Code" — highly recomm...	Public	2026-06-15 18:23:52
5	bilal_q	Bilal Qureshi	Working on my first startup idea. Exciting times!	Public	2026-06-15 18:23:52
12	bilal_q	Bilal Qureshi	New AI tools are changing everything in tech	Public	2026-06-15 18:23:52
13	nida_f	Nida Farooq	Poem I wrote last night... "Stars above Lahore..."	Friends	2026-06-15 18:23:52

Result 1 x

Output

Action Output

#	Time	Action	Message
25	18:24:30	INSERT INTO Notifications (notification_type, reference_id, is_read, user_id, triggered_by) VALUES (...)	15 row(s) affected Records: 15 Duplicates: 0 Warnings: 0
26	18:25:04	SELECT p.post_id, u.username, u.full_name, p.content, p.privacy, p.created_at FROM Posts p INNER JOIN Users u ON p.user_id = u.user_id ORDER BY p.created_at DESC;	15 row(s) returned

1.2 (b) LEFT JOIN — All users and their posts (includes users with no posts)

```
298 -- (b) LEFT JOIN — All users and their posts (includes users with no posts)
299 • SELECT u.user_id, u.username, p.post_id, p.content
300 FROM Users u
301 LEFT JOIN Posts p ON u.user_id = p.user_id
302 ORDER BY u.user_id;
303
```

user_id	username	post_id	content
1	hussain_d	1	Just finished implementing BFS in C++
1	hussain_d	9	DSA assignment submitted on time for once
2	talha_k	6	Just deployed my portfolio website. Check it out!
2	talha_k	14	Anyone else obsessed with Notion for productiv...
3	ali_h	7	Missing home-cooked food ☹️ Lahore ki biryani ...
4	sara_m	2	Beautiful sunset from Lahore today 🌅
4	sara_m	8	Photography tip: golden hour is everything 📸
5	haseeb_r	3	New personal best at the gym 💪
5	haseeb_r	15	Just hit 500 followers! Thank you all 🙏
6	zara_a	11	Eid Mubarak to everyone! 🌙
7	usman_b	10	Anime recommendation: Vinland Saga — master...
8	aisha_n	4	Started reading "Clean Code" — highly recomm...
9	bilal_q	5	Working on my first startup idea. Exciting times!
9	bilal_q	12	New AI tools are changing everything in tech
10	nida_f	13	Poem I wrote last night... "Stars above Lahore..."

Result 2 x

Output

Action Output

#	Time	Action	Message
26	18:25:04	SELECT p.post_id, u.username, u.full_name, p.content, p.privacy, p.created_at FROM Posts p INNER JOIN Users u ON p.user_id = u.user_id ORDER BY p.created_at DESC;	15 row(s) returned
27	18:27:32	SELECT u.user_id, u.username, p.post_id, p.content FROM Users u LEFT JOIN Posts p ON u.user_id = p.user_id ORDER BY u.user_id;	15 row(s) returned

1.3 (c) RIGHT JOIN — All posts and their comments (includes posts with no comments)

```
304 -- (c) RIGHT JOIN — All posts and their comments (include posts with no comments)
305 • SELECT p.post_id, p.content, c.comment_id, c.comment_text
306 FROM Comments c
307 RIGHT JOIN Posts p ON c.post_id = p.post_id
308 ORDER BY p.post_id;
309
```

post_id	content	comment_id	comment_text
1	Just finished implementing BFS in C++ 🤖	1	Great work Hussain! BFS is tough
1	Just finished implementing BFS in C++ 🤖	2	Thanks Talha! It took me hours 🤔
2	Beautiful sunset from Lahore today 🌅	5	Love this photo Sara!
2	Beautiful sunset from Lahore today 🌅	6	Thank you Zara! 🙏
3	New personal best at the gym 💪	3	Which gym do you go to?
3	New personal best at the gym 💪	4	Power House Gym — best in Lahore!
4	Started reading "Clean Code" — highly recomm...	9	I am also reading Clean Code!
4	Started reading "Clean Code" — highly recomm...	10	Which chapter are you on?
5	Working on my first startup idea. Exciting times!	14	Starting my startup too! DM me
6	Just deployed my portfolio website. Check it out!	NULL	NULL
7	Missing home-cooked food 🍲 Lahore ki biryani ...	NULL	NULL
8	Photography tip: golden hour is everything 📸	NULL	NULL
9	DSA assignment submitted on time for once 🙏	NULL	NULL
10	Anime recommendation: Vinland Saga — master...	7	Vinland Saga is goats 🐐
10	Anime recommendation: Vinland Saga — master...	8	Season 2 is even better!
11	Eid Mubarak to everyone! 🌙	NULL	NULL
12	New AI tools are changing everything in tech	NULL	NULL
13	Poem I wrote last night... "Stars above Lahore..."	15	This poem is beautiful Nida!
14	Anyone else obsessed with Notion for productiv...	13	Notion is life-changing!
15	Just hit 500 followers! Thank you all 🙏	11	Congratulations on 500!
15	Just hit 500 followers! Thank you all 🙏	12	Love your content Haseeb!

Result 3 x

Output

Action Output

#	Time	Action	Message
27	18:27:32	SELECT u.user_id, u.username, p.post_id, p.content FROM Users u LEFT JOIN Posts p ON u.user_id = p.user_id	15 row(s) returned

1.4 (d) FULL OUTER JOIN (emulated with UNION)

```
310 -- (d) FULL OUTER JOIN (emulated with UNION) -
311 -- All users + all posts even if no match on either side
312 • SELECT u.username, p.post_id, p.content
313 FROM Users u LEFT JOIN Posts p ON u.user_id = p.user_id
314 UNION
315 SELECT u.username, p.post_id, p.content
316 FROM Users u RIGHT JOIN Posts p ON u.user_id = p.user_id;
317
```

post_id	content	comment_id	comment_text
1	Just finished implementing BFS in C++ 🤖	1	Great work Hussain! BFS is tough
1	Just finished implementing BFS in C++ 🤖	2	Thanks Talha! It took me hours 🤔
2	Beautiful sunset from Lahore today 🌅	5	Love this photo Sara!
2	Beautiful sunset from Lahore today 🌅	6	Thank you Zara! 🙏
3	New personal best at the gym 💪	3	Which gym do you go to?
3	New personal best at the gym 💪	4	Power House Gym — best in Lahore!
4	Started reading "Clean Code" — highly recomm...	9	I am also reading Clean Code!
4	Started reading "Clean Code" — highly recomm...	10	Which chapter are you on?
5	Working on my first startup idea. Exciting times!	14	Starting my startup too! DM me
6	Just deployed my portfolio website. Check it out!	NULL	NULL
7	Missing home-cooked food 🍲 Lahore ki biryani ...	NULL	NULL
8	Photography tip: golden hour is everything 📸	NULL	NULL
9	DSA assignment submitted on time for once 🙏	NULL	NULL
10	Anime recommendation: Vinland Saga — master...	7	Vinland Saga is goats 🐐
10	Anime recommendation: Vinland Saga — master...	8	Season 2 is even better!
11	Eid Mubarak to everyone! 🌙	NULL	NULL
12	New AI tools are changing everything in tech	NULL	NULL
13	Poem I wrote last night... "Stars above Lahore..."	15	This poem is beautiful Nida!
14	Anyone else obsessed with Notion for productiv...	13	Notion is life-changing!
15	Just hit 500 followers! Thank you all 🙏	11	Congratulations on 500!
15	Just hit 500 followers! Thank you all 🙏	12	Love your content Haseeb!

Result 4 x

Output

Action Output

#	Time	Action	Message
17	18:23:52	INSERT INTO Posts (content, privacy, user_id) VALUES (Just finished implementing BFS in C++ 🤖, '...', 15 row(s) affected Records: 15 Duplicates: 0 Warnings: 0	

1.5 (e) Multi-table JOIN — Comments with post content and commenter username

```
318 -- (e) Multi-table JOIN - Comments with post content and commenter username
319 • SELECT c.comment_id,
320        u.username AS commenter,
321        p.content AS on_post,
322        c.comment_text,
323        c.created_at
324 FROM Comments c
325 JOIN Users u ON c.user_id = u.user_id
326 JOIN Posts p ON c.post_id = p.post_id
327 ORDER BY c.created_at;
```

comment_id	commenter	on_post	comment_text	created_at
10	aisha_n	Started reading "Clean Code" — highly recomm...	Which chapter are you on?	2026-06-15 18:24:03
15	aisha_n	Poem I wrote last night... "Stars above Lahore..."	This poem is beautiful Nida!	2026-06-15 18:24:03
8	ali_h	Anime recommendation: Vinland Saga — master...	Season 2 is even better!	2026-06-15 18:24:03
14	ali_h	Working on my first startup idea. Exciting times!	Starting my startup too! DM me	2026-06-15 18:24:03
4	haseeb_r	New personal best at the gym 🏋️	Power House Gym — best in Lahore!	2026-06-15 18:24:03
13	haseeb_r	Anyone else obsessed with Notion for productiv...	Notion is life-changing!	2026-06-15 18:24:03
2	hussain_d	Just finished implementing BFS in C++ 🐼	Thanks Talha! It took me hours 🙏	2026-06-15 18:24:03
7	hussain_d	Anime recommendation: Vinland Saga — master...	Vinland Saga is goated 🐼	2026-06-15 18:24:03
11	hussain_d	Just hit 500 followers! Thank you all 🙏	Congratulations on 500!	2026-06-15 18:24:03
6	sara_m	Beautiful sunset from Lahore today 🌅	Thank you Zara! 🙏	2026-06-15 18:24:03
12	sara_m	Just hit 500 followers! Thank you all 🙏	Love your content Haseeb!	2026-06-15 18:24:03
1	talha_k	Just finished implementing BFS in C++ 🐼	Great work Hussain! BFS is tough	2026-06-15 18:24:03
9	talha_k	Started reading "Clean Code" — highly recomm...	I am also reading Clean Code!	2026-06-15 18:24:03
3	usman_b	New personal best at the gym 🏋️	Which gym do you go to?	2026-06-15 18:24:03
5	zara_a	Beautiful sunset from Lahore today 🌅	Love this photo Sara!	2026-06-15 18:24:03

Result 5 ×

Output

Action Output

#	Time	Action	Message
✓ 29	18:29:08	SELECT p.post_id, p.content, c.comment_id, c.comment_text FROM Comments c RIGHT JOIN Post...	21 row(s) returned

1.6 (f) Self JOIN via Followers — Who follows whom (names instead of IDs)

```
328
329 -- (f) Self JOIN via Followers - Who follows whom (names instead of IDs)
330 • SELECT f1.username AS follower, f2.username AS following
331 FROM Followers f1
332 JOIN Users f1 ON f1.follower_id = f1.user_id
333 JOIN Users f2 ON f1.following_id = f2.user_id
334 ORDER BY f1.username;
```

follower	following
aisha_n	sara_m
aisha_n	bilal_q
aisha_n	nida_f
ali_h	hussain_d
ali_h	talha_k
ali_h	usman_b
bilal_q	hussain_d
bilal_q	talha_k
haseeb_r	hussain_d
haseeb_r	talha_k
haseeb_r	sara_m
hussain_d	talha_k
hussain_d	sara_m
hussain_d	haseeb_r
hussain_d	bilal_q
nida_f	sara_m
nida_f	zara_a
nida_f	aisha_n
sara_m	haseeb_r

Result 7 ×

Output

Action Output

#	Time	Action	Message
✓ 31	18:31:46	SELECT f1.username AS follower, f2.username AS following FROM Followers f1 JOIN Users f1 ON f1.f...	29 row(s) returned
✓ 32	18:32:13	SELECT f1.username AS follower, f2.username AS following FROM Followers f1 JOIN Users f1 ON f1.f...	29 row(s) returned

02

AGGREGATE FUNCTIONS

3 queries in this section

2.1 (a) COUNT, MAX, MIN, AVG, SUM — Overall platform stats

```
337 --
338 -- 3.2 AGGREGATE FUNCTIONS
339 --
340
341 -- (a) COUNT, MAX, MIN, AVG, SUM — Overall platform stats
342 • SELECT
343     COUNT(DISTINCT u.user_id) AS total_users,
344     COUNT(DISTINCT p.post_id) AS total_posts
345 FROM Users u, Posts p
346
```

	total_users	total_posts
▶	10	15

Result 10 ×

Output

Action Output

#	Time	Action	Message
37	18:35:57	SELECT COUNT(DISTINCT u.user_id) AS total_users, COUNT(DISTINCT p.post_id) AS tot...	1 row(s) returned
38	18:35:59	SELECT COUNT(DISTINCT u.user_id) AS total_users, COUNT(DISTINCT p.post_id) AS tot...	1 row(s) returned

```
347 • SELECT
348     COUNT(*) AS total_likes,
349     MAX(liked_at) AS most_recent_like,
350     MIN(liked_at) AS oldest_like
351 FROM Post_Likes;
352
```

	total_likes	most_recent_like	oldest_like
▶	75	2026-06-15 18:24:22	2026-06-15 18:24:22

Result 11 ×

Output

Action Output

#	Time	Action	Message
38	18:35:59	SELECT COUNT(DISTINCT u.user_id) AS total_users, COUNT(DISTINCT p.post_id) AS tot...	1 row(s) returned
39	18:36:26	SELECT COUNT(*) AS total_likes, MAX(liked_at) AS most_recent_like, M...	1 row(s) returned

2.2 (b) Likes per post (aggregate on junction table)

```
353 -- (b) Likes per post (aggregate on junction table)
354 • SELECT post_id, COUNT(*) AS like_count
355 FROM Post_Likes
356 GROUP BY post_id
357 ORDER BY like_count DESC;
358
```

Result Grid

	post_id	like_count
▶	11	10
	2	6
	8	6
	1	5
	5	5
	10	5
	12	5
	13	5
	14	5
	15	5
	3	4
	4	4
	6	4

Result 12 ×

Output

Action Output

#	Time	Action	Message
✓ 39	18:36:26	SELECT COUNT(*) AS total_likes, MAX(liked_at) AS most_recent_like, M...	1 row(s) returned
✓ 40	18:36:53	SELECT post_id, COUNT(*) AS like_count FROM Post_Likes GROUP BY post_id ORDER BY like_c...	15 row(s) returned

2.3 (c) Average comments per post

```
359 -- (c) Average comments per post
360 • SELECT AVG(comment_count) AS avg_comments_per_post
361 FROM (
362     SELECT post_id, COUNT(*) AS comment_count
363     FROM Comments
364     GROUP BY post_id
365 ) AS cc;
366
```

Result Grid

	avg_comments_per_post
▶	1.6667

Result 13 ×

Output

Action Output

#	Time	Action	Message
✓ 40	18:36:53	SELECT post_id, COUNT(*) AS like_count FROM Post_Likes GROUP BY post_id ORDER BY like_c...	15 row(s) returned
✓ 41	18:37:21	SELECT AVG(comment_count) AS avg_comments_per_post FROM (SELECT post_id, COUNT(*)...	1 row(s) returned

03

GROUP BY + HAVING

8 queries in this section

3.1 (a) GROUP BY — Number of posts per user

```
368 -- 
369 -- 3.3 GROUP BY + HAVING
370 -- 
371 
372 -- (a) GROUP BY — Number of posts per user
373 • SELECT u.username, COUNT(p.post_id) AS post_count
374 FROM Users u
375 LEFT JOIN Posts p ON u.user_id = p.user_id
376 GROUP BY u.user_id, u.username
377 ORDER BY post_count DESC;
378
```

username	post_count
bilal_q	2
haseeb_r	2
hussain_d	2
sara_m	2
talha_k	2
aisha_n	1
ali_h	1
nida_f	1
usman_b	1
zara_a	1

Result 14 x

Output

Action Output

#	Time	Action	Message
41	18:37:21	SELECT AVG(comment_count) AS avg_comments_per_post FROM (SELECT post_id, COUNT(*)	1 row(s) returned
42	18:37:47	SELECT u.username, COUNT(p.post_id) AS post_count FROM Users u LEFT JOIN Posts p ON u.us...	10 row(s) returned

3.2 (b) HAVING — Users who have more than 1 post

```
379 -- (b) HAVING — Users who have more than 1 post
380 • SELECT u.username, COUNT(p.post_id) AS post_count
381 FROM Users u
382 LEFT JOIN Posts p ON u.user_id = p.user_id
383 GROUP BY u.user_id, u.username
384 HAVING post_count > 1
385 ORDER BY post_count DESC;
386
```

username	post_count
bilal_q	2
haseeb_r	2
hussain_d	2
sara_m	2
talha_k	2

Result 15 x

Output

Action Output

#	Time	Action	Message
42	18:37:47	SELECT u.username, COUNT(p.post_id) AS post_count FROM Users u LEFT JOIN Posts p ON u.us...	10 row(s) returned
43	18:38:50	SELECT u.username, COUNT(p.post_id) AS post_count FROM Users u LEFT JOIN Posts p ON u.us...	5 row(s) returned

3.3 (c) GROUP BY — Total likes received per user

```
387 -- (c) GROUP BY — Total likes received per user
388 • SELECT u.username, COUNT(pl.post_id) AS total_likes_received
389 FROM Users u
390 JOIN Posts p ON p.user_id = u.user_id
391 JOIN Post_Likes pl ON pl.post_id = p.post_id
392 GROUP BY u.user_id, u.username
393 ORDER BY total_likes_received DESC;
394
```

Result Grid

username	total_likes_received
sara_m	12
bilal_q	10
zara_a	10
haseeb_r	9
talha_k	9
hussain_d	8
nida_f	5
usman_b	5
aisha_n	4
ali_h	3

Result 16 ×

Output

Action Output

#	Time	Action	Message
43	18:38:50	SELECT u.username, COUNT(p.post_id) AS post_count FROM Users u LEFT JOIN Posts p ON u.us...	5 row(s) returned
44	18:39:08	SELECT u.username, COUNT(pl.post_id) AS total_likes_received FROM Users u JOIN Posts p ON...	10 row(s) returned

3.4 (d) HAVING — Users whose posts collectively received more than 5 likes

```
395 -- (d) HAVING — Users whose posts collectively received more than 5 likes
396 • SELECT u.username, COUNT(pl.post_id) AS total_likes_received
397 FROM Users u
398 JOIN Posts p ON p.user_id = u.user_id
399 JOIN Post_Likes pl ON pl.post_id = p.post_id
400 GROUP BY u.user_id, u.username
401 HAVING total_likes_received > 5
402 ORDER BY total_likes_received DESC;
403
```

Result Grid

username	total_likes_received
sara_m	12
bilal_q	10
zara_a	10
haseeb_r	9
talha_k	9
hussain_d	8

Result 17 ×

Output

Action Output

#	Time	Action	Message
44	18:39:08	SELECT u.username, COUNT(pl.post_id) AS total_likes_received FROM Users u JOIN Posts p ON...	10 row(s) returned
45	18:39:23	SELECT u.username, COUNT(pl.post_id) AS total_likes_received FROM Users u JOIN Posts p ON...	6 row(s) returned

3.5 (e) GROUP BY — Follower count per user

```
404 -- (e) GROUP BY — Follower count per user
405 • SELECT u.username, COUNT(f.follower_id) AS followers
406 FROM Users u
407 LEFT JOIN Followers f ON f.following_id = u.user_id
408 GROUP BY u.user_id, u.username
409 ORDER BY followers DESC;
410
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

username	followers
hussain_d	5
sara_m	5
talha_k	4
bilal_q	3
nida_f	3
aisha_n	2
ali_h	2
haseeb_r	2
zara_a	2
usman_b	1

Result 18 x

Output

Action Output

#	Time	Action	Message
45	18:39:23	SELECT u.username, COUNT(pl.post_id) AS total_likes_received FROM Users u JOIN Posts p ON...	6 row(s) returned
46	18:39:44	SELECT u.username, COUNT(f.follower_id) AS followers FROM Users u LEFT JOIN Followers f ON f....	10 row(s) returned

3.6 (f) HAVING — Users with more than 3 followers

```
411 -- (f) HAVING — Users with more than 3 followers
412 • SELECT u.username, COUNT(f.follower_id) AS followers
413 FROM Users u
414 LEFT JOIN Followers f ON f.following_id = u.user_id
415 GROUP BY u.user_id, u.username
416 HAVING followers > 3
417 ORDER BY followers DESC;
418
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

username	followers
hussain_d	5
sara_m	5
talha_k	4

Result 19 x

Output

Action Output

#	Time	Action	Message
46	18:39:44	SELECT u.username, COUNT(f.follower_id) AS followers FROM Users u LEFT JOIN Followers f ON f....	10 row(s) returned
47	18:40:01	SELECT u.username, COUNT(f.follower_id) AS followers FROM Users u LEFT JOIN Followers f ON f....	3 row(s) returned

3.7 (g) Notifications grouped by type

```
419  -- (g) Notifications grouped by type
420  • SELECT notification_type, COUNT(*) AS count
421  FROM Notifications
422  GROUP BY notification_type;
423
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [↗](#)

notification_type	count
Like	6
Comment	4
Follow	4
Message	1

Result 20 ×

Output

Action Output

#	Time	Action	Message
✓ 47	18:40:01	SELECT u.username, COUNT(f.follower_id) AS followers FROM Users u LEFT JOIN Followers f ON f...	3 row(s) returned
✓ 48	18:40:13	SELECT notification_type, COUNT(*) AS count FROM Notifications GROUP BY notification_type LI...	4 row(s) returned

3.8 (h) Unread notifications per user (HAVING)

```
424  -- (h) Unread notifications per user (HAVING)
425  • SELECT u.username, COUNT(n.notification_id) AS unread_count
426  FROM Users u
427  JOIN Notifications n ON n.user_id = u.user_id
428  WHERE n.is_read = FALSE
429  GROUP BY u.user_id, u.username
430  HAVING unread_count >= 1
431  ORDER BY unread_count DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [↗](#)

username	unread_count
sara_m	2
usman_b	2
nida_f	2
hussain_d	1
bilal_q	1
haseeb_r	1

Result 21 ×

Output

Action Output

#	Time	Action	Message
✓ 48	18:40:13	SELECT notification_type, COUNT(*) AS count FROM Notifications GROUP BY notification_type LI...	4 row(s) returned
✓ 49	18:40:43	SELECT u.username, COUNT(n.notification_id) AS unread_count FROM Users u JOIN Notifications ...	6 row(s) returned

04

SUBQUERIES (NON-CORRELATED)

5 queries in this section

4.1 (a) Users who have never posted

```
438 -- (a) Users who have never posted
439 • SELECT username, full_name
440 FROM Users
441 WHERE user_id NOT IN (SELECT DISTINCT user_id FROM Posts);
442
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

username	full_name
----------	-----------

Users 23 x

Output

Action Output

#	Time	Action	Message
50	18:41:23	SELECT username, full_name FROM Users WHERE user_id NOT IN (SELECT DISTINCT user_id F...	0 row(s) returned
51	18:41:25	SELECT username, full_name FROM Users WHERE user_id NOT IN (SELECT DISTINCT user_id F...	0 row(s) returned

4.2 (b) The most liked post (single scalar subquery)

```
443 -- (b) The most liked post (single scalar subquery)
444 • SELECT post_id, content
445 FROM Posts
446 WHERE post_id = (
447     SELECT post_id
448     FROM Post_Likes
449     GROUP BY post_id
450     ORDER BY COUNT(*) DESC
451     LIMIT 1
452 )
453
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

post_id	content
11	Eid Mubarak to everyone! 🌙
NULL	NULL

Posts 24 x

Output

Action Output

#	Time	Action	Message
51	18:41:25	SELECT username, full_name FROM Users WHERE user_id NOT IN (SELECT DISTINCT user_id F...	0 row(s) returned
52	18:41:42	SELECT post_id, content FROM Posts WHERE post_id = (SELECT post_id FROM Post_Likes ...	1 row(s) returned

4.3 (c) Posts that have at least one comment

```
454 -- (c) Posts that have at least one comment
455 • SELECT post_id, content
456 FROM Posts
457 WHERE post_id IN (SELECT DISTINCT post_id FROM Comments);
458
```

Result Grid

post_id	content
1	Just finished implementing BFS in C++ 🐼
2	Beautiful sunset from Lahore today 🌅
3	New personal best at the gym 💪
4	Started reading "Clean Code" — highly recomm...
5	Working on my first startup idea. Exciting times!
10	Anime recommendation: Vinland Saga — master...
13	Poem I wrote last night... "Stars above Lahore..."
14	Anyone else obsessed with Notion for productiv...
15	Just hit 500 followers! Thank you all ❤️
NULL	NULL

Posts 25 ×

Output

Action Output

#	Time	Action	Message
✓ 52	18:41:42	SELECT post_id, content FROM Posts WHERE post_id = (SELECT post_id FROM Post_Likes ...	1 row(s) returned
✓ 53	18:41:57	SELECT post_id, content FROM Posts WHERE post_id IN (SELECT DISTINCT post_id FROM Com...	9 row(s) returned

4.4 (d) Users who follow the most-followed user

```
459 -- (d) Users who follow the most-followed user
460 • SELECT username FROM Users
461 WHERE user_id IN (
462     SELECT follower_id FROM Followers
463     WHERE following_id = (
464         SELECT following_id FROM Followers
465         GROUP BY following_id
466         ORDER BY COUNT(*) DESC
467         LIMIT 1
468     )
469 )
```

Result Grid

username
talha_k
ali_h
haseeb_r
usman_b
bilal_q

Users 26 ×

Output

Action Output

#	Time	Action	Message
✓ 53	18:41:57	SELECT post_id, content FROM Posts WHERE post_id IN (SELECT DISTINCT post_id FROM Com...	9 row(s) returned
✓ 54	18:42:39	SELECT username FROM Users WHERE user_id IN (SELECT follower_id FROM Followers W...	5 row(s) returned

4.5 (e) Posts with above-average likes

```
471 -- (e) Posts with above-average likes
472 • SELECT p.post_id, p.content, COUNT(pl.user_id) AS likes
473 FROM Posts p
474 JOIN Post_Likes pl ON pl.post_id = p.post_id
475 GROUP BY p.post_id, p.content
476 HAVING likes > (
477     SELECT AVG(like_count)
478     FROM (
479         SELECT COUNT(*) AS like_count FROM Post_Likes
480         GROUP BY post_id
481     ) AS avg_table
482 );
483
```

Result Grid

	post_id	content	likes
▶	2	Beautiful sunset from Lahore today 🌄	6
	8	Photography tip: golden hour is everything 📸	6
	11	Eid Mubarak to everyone! 🌙	10

Result 27 ×

Output

Action Output

#	Time	Action	Message
✓ 54	18:42:39	SELECT username FROM Users WHERE user_id IN (SELECT follower_id FROM Followers W...	5 row(s) returned
✓ 55	18:43:08	SELECT p.post_id, p.content, COUNT(pl.user_id) AS likes FROM Posts p JOIN Post_Likes pl ON pl....	3 row(s) returned

05

CORRELATED SUBQUERIES

5.1 (a) Users who have liked every post they commented on

```
489 -- (a) Users who have liked every post they commented on
490 • SELECT u.username FROM Users u
491 WHERE NOT EXISTS (
492     SELECT c.post_id FROM Comments c
493     WHERE c.user_id = u.user_id
494     AND c.post_id NOT IN (
495         SELECT pl.post_id FROM Post_Likes pl
496         WHERE pl.user_id = u.user_id
497     )
498 )
```

Result Grid

username
bilal_q
nida_f
talha_k
zara_a

Users 28 x

Output

Action Output

#	Time	Action	Message
55	18:43:08	SELECT p.post_id, p.content, COUNT(pl.user_id) AS likes FROM Posts p JOIN Post_Likes pl ON p...	3 row(s) returned
56	18:43:49	SELECT u.username FROM Users u WHERE NOT EXISTS (SELECT c.post_id FROM Comment...	4 row(s) returned

5.2 (b) Each user's most recent post (correlated in FROM / lateral-style using MAX)

```
500 -- (b) Each user's most recent post (correlated in FROM / lateral-style using MAX)
501 • SELECT u.username,
502     p.post_id,
503     p.content,
504     p.created_at
505 FROM Posts p
506 JOIN Users u ON p.user_id = u.user_id
507 WHERE p.created_at = (
508     SELECT MAX(p2.created_at)
509     FROM Posts p2
510     WHERE p2.user_id = p.user_id -- correlated on outer table
511 )
```

Result Grid

username	post_id	content	created_at
aisha_n	4	Started reading "Clean Code" — highly recomm...	2026-06-15 18:23:52
ali_h	7	Missing home-cooked food ☹️ Lahore ki biryani ...	2026-06-15 18:23:52
bilal_q	5	Working on my first startup idea. Exciting times!	2026-06-15 18:23:52
bilal_q	12	New AI tools are changing everything in tech	2026-06-15 18:23:52
haseeb_r	3	New personal best at the gym 🏋️	2026-06-15 18:23:52
haseeb_r	15	Just hit 500 followers! Thank you all 🙏	2026-06-15 18:23:52
hussain_d	1	Just finished implementing BFS in C++ 🐞	2026-06-15 18:23:52
hussain_d	9	DSA assignment submitted on time for once 🙏	2026-06-15 18:23:52
nida_f	13	Poem I wrote last night... "Stars above Lahore..."	2026-06-15 18:23:52
sara_m	2	Beautiful sunset from Lahore today 🌅	2026-06-15 18:23:52
sara_m	8	Photography tip: golden hour is everything 📸	2026-06-15 18:23:52

Result 29 x

Output

Action Output

#	Time	Action	Message
56	18:43:49	SELECT u.username FROM Users u WHERE NOT EXISTS (SELECT c.post_id FROM Comment...	4 row(s) returned
57	18:44:07	SELECT u.username, p.post_id, p.content, p.created_at FROM Posts p JOIN Users u ...	15 row(s) returned

5.3 (c) Posts that received more likes than the author's own average

```
513 -- (c) Posts that received more likes than the author's own average
514 • SELECT p.post_id, p.content,
515       (SELECT COUNT(*) FROM Post_Likes pl WHERE pl.post_id = p.post_id) AS likes
516 FROM Posts p
517 WHERE (SELECT COUNT(*) FROM Post_Likes pl WHERE pl.post_id = p.post_id)
518       >
519       (SELECT AVG(cnt) FROM (
520         SELECT COUNT(*) AS cnt FROM Post_Likes pl2
521         JOIN Posts p3 ON pl2.post_id = p3.post_id
522         WHERE p3.user_id = p.user_id -- correlated to outer Posts
523         GROUP BY pl2.post_id
524       ) AS author_avg);
525
```

Result Grid | Filter Rows: | Exports: | Wrap Cell Contents:

	post_id	content	likes
▶	1	Just finished implementing BFS in C++ 🐼	5
	14	Anyone else obsessed with Notion for productiv...	5
	15	Just hit 500 followers! Thank you all ❤️	5

Result 30 ×

Output

Action Output

#	Time	Action	Message
✓	57 18:44:07	SELECT u.username, p.post_id, p.content, p.created_at FROM Posts p JOIN Users u ...	15 row(s) returned
✓	58 18:44:51	SELECT p.post_id, p.content, (SELECT COUNT(*) FROM Post_Likes pl WHERE pl.post_id = p....	3 row(s) returned

5.4 (d) Users who have sent messages but never received any (EXISTS / NOT EXISTS)

```
527 • SELECT u.username
528 FROM Users u
529 WHERE EXISTS (SELECT 1 FROM Messages m WHERE m.sender_id = u.user_id)
530 AND EXISTS (SELECT 1 FROM Messages m WHERE m.receiver_id = u.user_id);
```

Result Grid

username
hussain_d
talha_k
ali_h
sara_m
haseeb_r
zara_a
usman_b
bilal_q

Users 32 x

Output

#	Time	Action	Message
59	18:45:09	SELECT u.username FROM Users u WHERE EXISTS (SELECT 1 FROM Messages m WHERE m...	0 row(s) returned
60	18:45:22	SELECT u.username FROM Users u WHERE EXISTS (SELECT 1 FROM Messages m WHERE m...	8 row(s) returned

```
526 -- (d) Users who have sent messages but never received any (EXISTS / NOT EXISTS)
527 • SELECT u.username
528 FROM Users u
529 WHERE EXISTS (SELECT 1 FROM Messages m WHERE m.sender_id = u.user_id)
530 AND NOT EXISTS (SELECT 1 FROM Messages m WHERE m.receiver_id = u.user_id);
```

Result Grid

username

Users 33 x

Output

#	Time	Action	Message
60	18:45:22	SELECT u.username FROM Users u WHERE EXISTS (SELECT 1 FROM Messages m WHERE m...	8 row(s) returned
61	18:45:36	SELECT u.username FROM Users u WHERE EXISTS (SELECT 1 FROM Messages m WHERE m...	0 row(s) returned

5.5 (e) Comments that received more likes than the average comment-like count

```
532 -- (e) Comments that received more likes than the average comment-like count
533 • SELECT c.comment_id, c.comment_text,
534       (SELECT COUNT(*) FROM Comment_Likes cl WHERE cl.comment_id = c.comment_id) AS likes
535 FROM Comments c
536 WHERE (SELECT COUNT(*) FROM Comment_Likes cl WHERE cl.comment_id = c.comment_id)
537       >
538       (SELECT AVG(lc) FROM (
539         SELECT COUNT(*) AS lc
540         FROM Comment_Likes
541         GROUP BY comment_id
542       ) AS avg_cl);
543
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [T](#)

comment_id	comment_text	likes
7	Vinland Saga is goated 🐉	4
11	Congratulations on 500!	4

Result 34 x

Output

Action Output

#	Time	Action	Message
61	18:45:36	SELECT u.username FROM Users u WHERE EXISTS (SELECT 1 FROM Messages m WHERE m...	0 row(s) returned
62	18:45:53	SELECT c.comment_id, c.comment_text, (SELECT COUNT(*) FROM Comment_Likes cl WHER...	2 row(s) returned

06

SET OPERATIONS

5 queries in this section

6.1 (a) UNION — Users who have posted OR commented (distinct list)

```
545 -- =====
546 -- 3.6 SET OPERATIONS
547 -- =====
548
549 -- (a) UNION — Users who have posted OR commented (distinct list)
550 • SELECT user_id FROM Posts
551 UNION
552 SELECT user_id FROM Comments;
553
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

user_id
1
2
3
4
5
6
7
8
9
10

Result 35 x

Output

Action Output

#	Time	Action	Message
62	18:45:53	SELECT c.comment_id, c.comment_text, (SELECT COUNT(*) FROM Comment_Likes cl WHERE...	2 row(s) returned
63	18:46:24	SELECT user_id FROM Posts UNION SELECT user_id FROM Comments	10 row(s) returned

6.2 (b) UNION ALL — Every user interaction (posts + comments, with duplicates)

```
554 -- (b) UNION ALL — Every user interaction (posts + comments, with duplicates)
555 • SELECT user_id, 'Post' AS activity_type FROM Posts
556 UNION ALL
557 SELECT user_id, 'Comment' AS activity_type FROM Comments
558 ORDER BY user_id;
559
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

user_id	activity_type
1	Post
1	Post
1	Comment
1	Comment
1	Comment
2	Post
2	Post
2	Comment
2	Comment
3	Post
3	Comment

Result 36 x

Output

Action Output

#	Time	Action	Message
63	18:46:24	SELECT user_id FROM Posts UNION SELECT user_id FROM Comments	10 row(s) returned
64	18:46:39	SELECT user_id, 'Post' AS activity_type FROM Posts UNION ALL SELECT user_id, 'Comment' AS ...	30 row(s) returned

6.3 (c) INTERSECT (emulated) — Users who have BOTH posted AND commented

```
560 -- (c) INTERSECT (emulated) — Users who have BOTH posted AND commented
561 • SELECT user_id FROM Posts
562 WHERE user_id IN (SELECT user_id FROM Comments);
563
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	user_id
▶	1
	1
	2
	2
	3
	4
	4
	5
	5
	6
	7

Posts 37 x

Output

Action Output

#	Time	Action	Message
✓ 64	18:46:39	SELECT user_id, 'Post' AS activity_type FROM Posts UNION ALL SELECT user_id, 'Comment' AS ...	30 row(s) returned
✓ 65	18:46:58	SELECT user_id FROM Posts WHERE user_id IN (SELECT user_id FROM Comments) LIMIT 0, 1000	12 row(s) returned

6.4 (d) EXCEPT / MINUS (emulated) — Users who posted but have NOT commented

```
563
564 -- (d) EXCEPT / MINUS (emulated) — Users who posted but have NOT commented
565 • SELECT DISTINCT user_id FROM Posts
566 WHERE user_id NOT IN (SELECT DISTINCT user_id FROM Comments);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	user_id
▶	9
	10

Posts 38 x

Output

Action Output

#	Time	Action	Message
✓ 65	18:46:58	SELECT user_id FROM Posts WHERE user_id IN (SELECT user_id FROM Comments) LIMIT 0, 1000	12 row(s) returned
✓ 66	18:47:10	SELECT DISTINCT user_id FROM Posts WHERE user_id NOT IN (SELECT DISTINCT user_id FR...	2 row(s) returned

6.5 (e) Users who liked a post AND saved that same post (intersection via INTERSECT emulation)

```
568 -- (e) Users who liked a post AND saved that same post (intersection via INTERSECT emulation)
569 • SELECT user_id FROM Post_Likes
570 WHERE post_id = 1
571 AND user_id IN (
572     SELECT user_id FROM Saved_Posts WHERE post_id = 1
573 );
574
```

Result Grid

user_id
2
7

Post_Likes 39 x

Output

Action Output

#	Time	Action	Message
66	18:47:10	SELECT DISTINCT user_id FROM Posts WHERE user_id NOT IN (SELECT DISTINCT user_id FR...	2 row(s) returned
67	18:47:29	SELECT user_id FROM Post_Likes WHERE post_id = 1 AND user_id IN (SELECT user_id FR...	2 row(s) returned